Bhavethra Baabu

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Experience

Full Stack Developer Aug 2026 - Present
DQ Pursuit, Inc. *Salem, MA*

- Engineered and delivered **full-stack web application features** using **React, Next.js, TypeScript, Python, REST APIs, and PostgreSQL**, improving feature delivery speed by **30%**.
- Developed an **AI-powered chatbot** by integrating **Python services and REST APIs**, enabling natural-language queries for project details and onboarding workflows and reducing onboarding time by **50%**.
- Implemented and validated an **end-to-end Twilio Verify SMS authentication workflow** with **secure OTP handling, PostgreSQL persistence, audit logging, and security validation**, strengthening application authentication.
- Automated **build, testing, and validation workflows** by implementing **CI/CD pipelines**, reducing manual development overhead and accelerating application delivery cycles.

Software Engineer Intern - Backend & API Jan 2026 - May 2026
SciQuel *Allston, MA*

- Built and integrated **RESTful APIs** using **Next.js** and **TypeScript**, collaborating with backend services backed by **PostgreSQL** to deliver **production-ready features**.
- Investigated application performance bottlenecks and optimized **Prisma ORM** queries against **PostgreSQL**, improving **response times by 30%** under concurrent traffic.
- Developed reusable **React** components with **TypeScript** and **Tailwind CSS** following **atomic design principles**, improving maintainability and **accelerating feature delivery by 30%**.
- Resolved **production issues** across React, Next.js, and PostgreSQL, reducing customer-reported issues and improving application stability.

Software Engineer - Backend & Cloud Engineering Jul 2022 – Jul 2024
LTIMindtree *Bengaluru, India*

- Engineered backend modules for **Unitrax**, an enterprise financial management platform, using **Java**, **Spring Boot**, **Spring Data JPA**, and **Hibernate** to automate financial workflows and improve processing efficiency by **30%**.
- Developed secure **Managed File Transfer (MFT)** services on **Linux**, integrating enterprise financial systems through automated file processing and encrypted data exchange, reducing transfer failures by **25%**.
- Designed and constructed secure **REST APIs** with **Spring Boot**, integrating business logic, validation, exception handling, and RBAC to support scalable enterprise financial workflows and secure data processing.
- Refactored legacy backend services using **Spring Data JPA** and **Hibernate**, reducing code complexity by **30%**, improving query performance, and enhancing maintainability for faster feature delivery.

Projects

CampusPulse | GitHub | Live Demo *Next.js, TypeScript, Python, PostgreSQL, REST APIs, Qdrant* Jan 2026 – Apr 2026

- **Created CampusPulse**, an AI-powered academic search platform leveraging Qdrant and Groq Llama 3.1 to deliver natural language responses across 285+ university pages; presented at the **Clark University Tech Innovation Challenge.**
- Implemented **data ingestion, transformation, and indexing pipelines** to support real-time semantic search and AI-assisted content retrieval through REST APIs.

CoinPulse | GitHub | Live Demo *Next.js, TypeScript, PostgreSQL, WebSocket, REST APIs* Jan 2026 – Apr 2026

- Constructed a **WebSocket-based real-time Event-driven data pipeline** integrating Binance APIs to stream live market data and candlestick visualizations for **1,000+ cryptocurrency assets**.
- Optimized **server-side rendering and caching mechanisms**, delivering market updates within **2 seconds** while improving application responsiveness and data delivery performance.

Technical Skills

Languages: Java, Python, SQL, JavaScript, TypeScript
Frameworks & Frontend: Spring Boot, Spring Cloud Gateway, Spring Data JPA, FastAPI, React, Next.js, HTML5, CSS3
Backend: REST APIs, Microservices, API Gateway, JWT, OAuth2, JSON, HTTP/HTTPS, WebSockets
Databases & Messaging: PostgreSQL, MongoDB, Redis, pgvector, Kafka, Redis Streams, Amazon SQS
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch), Docker, Kubernetes, GitHub Actions, CI/CD, Maven
Testing & Observability: JUnit 5, Mockito, Postman, Prometheus, Grafana
Architecture: Distributed Systems, Caching, Load Balancing, Circuit Breaker, Microservices
AI/LLM: LangChain, RAG, Vector Search, OpenAI API, Prompt Engineering

Education

Clark University Worcester, MA
Master of Science in Computer Science, GPA: 3.8/4.0 *Aug 2024 – May 2026*

Panimalar Institute of Technology Chennai, India
Bachelor of Technology – Information Technology *Aug 2018 – Jul 2022*